

1: IP Address Classes

IP Class	IP Address Range	Default Subnet Mask	Purpose	Real-World Example
A	1.0.0.0 – 126.255.255.255	255.0.0.0 (/8)	Large networks	Google's large enterprise network
B	128.0.0.0 – 191.255.255.255	255.255.0.0 (/16)	Medium-sized organizations	A university campus network
C	192.0.0.0 – 223.255.255.255	255.255.255.0 (/24)	Small businesses and home networks	Home Wi-Fi router (e.g., 192.168.1.x)
D	224.0.0.0 – 239.255.255.255	N/A	Multicast communication	IPTV or live video streaming
E	240.0.0.0 – 255.255.255.255	N/A	Experimental/Research	Network protocol testing

2: Calculate Network Information

Given:

- IP Address: **192.168.10.45**
- Subnet Mask: **255.255.255.0 /24**

Network Address & Host Address

A /24 subnet means the first 24 bits are the network portion.

IP Address : 192.168.10.45

Subnet Mask : 255.255.255.0

Network	Host
$=2^n$	$=2^h - 2$
$=2^0$	$=2^8 - 2$
=1Network	=254 Host

Network Address : 192.168.10.0

Host Address : 192.168.10.1

Host Address : 192.168.10.254

Broadcast Address : 192.168.10.255

3: Subnetting Plan For Food Delivery

Given Network: 10.0.0.0/24

Need **4 subnets**.

$$2^2=4$$

Subnet Mask = 255.255.255.192

Network = 4

Host= 62

Department	Network Address	Host Range	Broadcast Address
Delivery Partners	10.0.0.0	10.0.0.1 – 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64	10.0.0.65 – 10.0.0.126	10.0.0.127
Customers	10.0.0.128	10.0.0.129 – 10.0.0.190	10.0.0.191
Admins	10.0.0.192	10.0.0.193 – 10.0.0.254	10.0.0.255

4: College Wi-Fi Subnetting

Given

Network:

172.16.0.0/16

Subnet mask 255.255.248.0

Need:

At least 20 subnets

$$2^5 = 32$$

Binary:

11111111.11111111.11111000.00000000

Network = 32

Host = 2046

Hosts per Subnet **2046 usable hosts**

5: Invalid IP Address

The IP address **192.168.1.300** is invalid because each octet in an IPv4 address must be between **0 and 255**. The value **300** exceeds the maximum allowed value of **255**, so the device cannot use this address and will not connect to the network.

For the subnet **192.168.1.0/24**, a valid host IP address should be in the range **192.168.1.1 to 192.168.1.254** (excluding **192.168.1.0**, which is the network address, and **192.168.1.255**, which is the broadcast address).

Examples of valid IP addresses:

- 192.168.1.10
- 192.168.1.50
- 192.168.1.100
- 192.168.1.200